

Real or Fake? Human Judgments about Photographs and Computer-generated Images of Faces

Shaojing Fan Tian-Tsong Ng* Jonathan S. Herberg Bryan L. Koenig Shiqing Xin
Ningbo University I²R, A*STAR IHPC, A*STAR IHPC, A*STAR Ningbo University

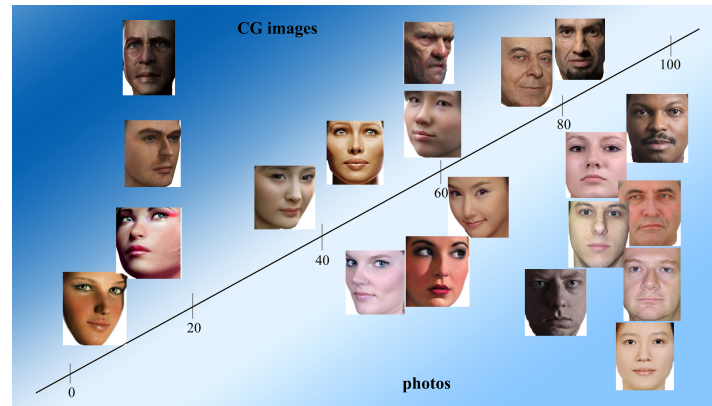


Figure 1: A psychophysical experiment explored factors underlying human judgments of visual realism for face images. The figure shows subjective realism score of 10 photo-realistic computer-generated (CG) images and 10 photos of faces. CG images are above the scale and photos below it. Overall, photos were judged to be more realistic than CG images.

Abstract

While efforts to create realistic images have generated breakthroughs in computer graphics modeling, there has been little research to date on factors causing people to perceive images as real versus computer-generated (CG). The motivation of the current experiment is to begin investigating such factors. We showed both computer-graphics experts and laypersons real photos and CG images. Photo and CG images were of three types: original, modified to show only intrinsic reflectance components, and modified to show only intrinsic shading components (grayscale). Participants judged whether each image was a real photo or a CG image. Results showed that visual realism depends not only on image properties, but also on cognitive characteristics of viewers. Shading was especially crucial for visual realism. Color was also important. Experts outperformed laypersons, but their advantage was limited to grayscale images. This research at the interface between human cognition and computer vision is a starting point for investigating the factors underlying visual realism.

CR Categories: I.3.7 [Computer Graphics]: Three-Dimensional Graphics and Realism—Color, shading, shadowing, and texture I.4.1 [Image Processing and Computer Vision]: Digitization and Image Capture—Reflectance J.4 [Social and Behavioral Sciences]: Psychology—;

Keywords: computer generated graphics, human perception, face perception, visual realism, evaluation

*Corresponding author. e-mail: ttng@i2r.a-star.edu.sg

Copyright © 2012 by the Association for Computing Machinery, Inc. Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers, or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions Dept, ACM Inc., fax +1 (212) 869-0481 or e-mail permissions@acm.org.

SIGGRAPH Asia 2012, Singapore, November 28 – December 1, 2012.
© 2012 ACM 978-1-4503-1757-3/12/0011\$15.00

1 Introduction

Determining how real an image appears - its *visual realism* - is important in the field of computer graphics. Unfortunately doing so is challenging, in part because factors influencing visual realism are not well understood. Knowledge of such factors would have implications for various fields. For example, it would help cognitive scientists to discover perceptual and cognitive mechanisms that underlie judgments of visual realism. Knowledge of factors influencing visual realism may also help computer-graphics designers to create more realistic visual media. It may also help in image forensics by improving computational models that distinguish CG images from photos. These benefits of understanding visual realism motivated our research.

Our experiments build on the research of multiple disciplines. The computer graphics community since the early 1980's has investigated *CG fidelity*, i.e., how visually real CG images appear as compared to real scenes [Meyer et al. 1986; McNamara 2005]. Image properties like illumination, shadow, and surface roughness were identified as important factors that influence CG fidelity. Recent work explored how different ways of rendering virtual humans relates to the Uncanny Valley phenomenon [McDonnell et al. 2012]. Image forensics researchers have been developing computational models to distinguish CG images from photos [Ng and Chang 2012]. However, these models were mainly based on low-level image features (e.g. wavelet coefficients) and provide little insight into what makes an image appear real or fake. Research in cognitive psychology and neuroscience supports the notion that human face perception is a holistic and configural process. Many researchers also argue it is domain-specific, whereas other researchers argue that processes and brain areas involved in face perception may be recruited for expertise-based perceptual categorization more generally, see [Bruce and Young 2012] for discussion.

Our work focuses on two factors largely unexplored for visual realism assessment. *First* are intrinsic image properties, specifically reflectance and shading. *Second* are individual differences among

judges in their image-processing expertise. To evaluate these factors, we performed a psychophysics experiment in which laypersons and image-processing experts viewed a series of photos and CG images of faces and judged whether they thought each was a photo or a CG image. Stimuli in our experiment were unaltered color images, images modified to show only a grayscale version, and images modified to show only a reflectance version. We analyzed judgment data using signal detection theory (a common analysis tool in psychophysics and biology) and inferential statistics. We interpreted the results from psychological and cognitive perspectives and identified tips for computer graphics designers. These experiments provide an early investigation into what visual and cognitive factors influence human perception of visual realism for images of faces.

2 Psychophysics Experiment

2.1 Participants

Participants were 122 experts and 261 laypersons. The numbers of female in each group were 51 and 104. All participants were 18 to 40 years old with normal or corrected-to-normal visual acuity and normal color vision. Most were Chinese. We recruited participants via email. We coded them as experts if they indicated using image processing software in their daily job. Many experts worked for animation corporations, advertising companies, architectural design centers, and universities where they taught advertising and digital multimedia or did research on CG rendering or exaggeration. Other participants were coded as laypersons. Our final sample of experts was 80 after we excluded the data of 42 who worked for animation corporations and therefore might have seen some of our CG images prior to participating in our study.

2.2 Stimulus Images

We collected a database of 500 pairs of images of faces. Each pair contained one CG image and one photograph. The faces in each pair were matched for age, gender, and race. All faces had a neutral expression. Most CG images were from a popular computer graphics website (www.forums.cgsociety.org). Most photographs were from a photo ranking website (www.photosig.com) or from open face recognition databases. We continue to expand the database by incorporating images from new Web sources.

For this experiment, we selected 10 pairs of photos and what appeared to us to be highly photo-realistic CG images from our face database (see Fig. 2(a)). We cropped them such that only the face is visible, with minimum hair and no clothes visible. Most images were down-sampled so that the area depicting the face was about 250×270 pixels. The image background was manually removed.



Figure 2: Sample images from (a) our own dataset and (b) Farid and Bravo's dataset [Farid and Bravo 2012].

2.3 Intrinsic Image Decomposition

We extracted shading and reflectance components from the face images through intrinsic image decomposition to produce images

that showed only the shading or reflectance components of the images. Our decomposition algorithm was a modification of that used in [Grosse et al. 2009]. We expanded the algorithm, which was originally designed for grayscale images, to three different color channels in order to make it applicable on color images. Shading accounts for brightness change due to the object geometry; it is similar to a grayscale version of the image. Our shading components were often blurry so we used the grayscale version of the images instead. Reflectance accounts for the color of the surface when 3D geometry effects are removed [Grosse et al. 2009]. Thus for each face image we had three types: original (orig), grayscale (gray), and reflectance (refl). We expected that CG images might be more "friendly" to our decomposition algorithm due to its distinctive stepwise generation process; however, intrinsic components of CG images and photos showed no obvious visual differences (see examples in Fig. 3).

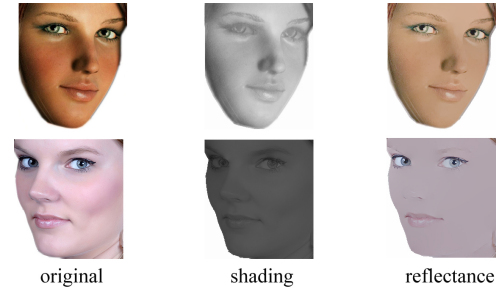


Figure 3: Two examples of intrinsic image decomposition, the above is a CG image, below is the photo in the pair.

2.4 Method

Participants completed the experiment over the Internet on a website that we created. After viewing instructions and examples, participants saw 60 images (20 images $\times$ 3 types). Each participant was randomly assigned to one of six counterbalanced orders of viewing the three image types. Within each image type, image presentation order was random for each participant. In response to a forced-choice question, participants judged each image as "CG" or "photo". Viewing time, viewing distance and display terminals were unconstrained. The website automatically recorded response time of each judgment. At the end of experiment, participants indicated what they believed they used as their main clue for distinguishing photos and CG images. They chose one option from the following list: eyes, skin, color, expression, illumination, lip, nose, others. Experimental sessions lasted approximately 20 minutes.

2.5 Results

Realism Score: We defined a realism score for an image as the percent of trials in which participants classified it as a photo (0-100) (see Fig. 1). CG images were distributed across the realism scale whereas photos concentrated near the high-realism end of the scale. The difference on distribution indicates that there may be something intrinsic to the realism of photos, despite their variations in compression ratio, resolution, retouching, and even semantics (gender, pose, expression). This probably suggests a possible alternative for studying visual realism—seeking the key factors in realism of photos, which remains less explored by researchers. Notably, four CG images were judged by over half of the participants as photos, indicating they seemed more like photos than CG images.

Signal Detection Theory: Signal detection theory (SDT) is a common analysis tool in psychophysics and biology [Wickens 2002]. It

offers a method of modeling the decision making process for someone who decides whether items are members of different classes (e.g., CG or photo). It also numerically expresses bias for when an observer favors a particular type of decision.

We define photo as the *signal* (class member) and CG as *noise* (not class member). Our SDT analysis includes 5 measures. In SDT, an observer's sensitivity, indexed by d' , is a measure of how well the observer can differentiate items coming from the Signal Absent and Signal Present distributions. Higher values of d' represent higher sensitivity. d' values near zero indicate chance performance. As this variable is estimated by the z -transformations of hit rates and false alarm rates, it requires in its calculation replacements of these rates when they are 0 or 1. Thus, rates of 0 were substituted with $1/2N$, and of 1 with $1 - (1/2N)$, where N is the number of items belonging to the class (for hit rates) or not belonging (for false alarm rates). An alternative to d' that makes fewer assumptions about the distribution of classification decisions is A' , which is a geometric approximation of the area under the *receiver operating characteristic* (ROC) curve. We also calculated the simple hit rate minus false alarm rate $hr\text{-}far$, which likewise gauges classification accuracy. Criterion (indexed by λ) represents the minimum level of internal certainty needed for the observer to decide that a signal was present. A larger value of λ indicates a bias for favoring CG responses. Another measure of response bias is $\log \beta$. A negative value of $\log \beta$ indicates a bias for favoring photo responses. The values of the first three measures averaged across participants are shown in Fig. 4. Detailed analyses are given below.

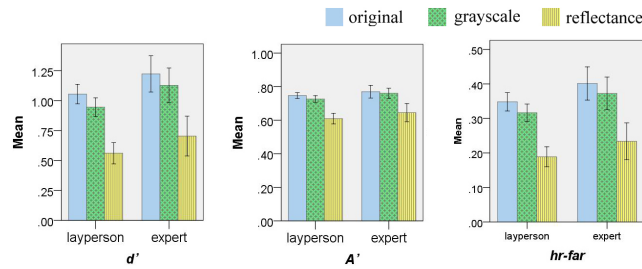


Figure 4: SDT values of three image types.

2.6 Data Analysis

Inferential Statistics: We conducted inferential statistical tests of differences in SDT measures. These tests are standard in behavioral and other sciences. See, for example, [Bailey 2008] for an introduction to inferential statistics. Our analysis strategy follows. Our outcome variables of interest were the SDT measures. The unit of analysis was each participant's mean SDT measures for each condition. We evaluated whether SDT measures differed by viewing order, image type, and participant characteristics (expertise-level and gender). In each case, we first perform multivariate analysis of variance (MANOVA) to check for general effects on SDT variables. If the MANOVA was significant, we then performed separate univariate analyses of variance (ANOVA's) on each SDT variable. If an ANOVA was significant, we followed up with pairwise comparisons of conditions using t -tests to see where differences were.

Viewing Order: We had no hypotheses regarding effects of viewing order, but nonetheless evaluate whether it influenced judgments. Recall that the display order for the original, grayscale and reflectance image types was random across participants. The MANOVA and all ANOVA's indicated no main effects of viewing order on any SDT measure, F 's(25, 1645) < .73, p 's > .817¹, but viewing order interacted with image type on λ , $F(9.5, 625.115) = 4.07$,

$p < .01$, and $\log \beta$, $F(10, 658) = 3.96$, $p < .01$. This means that the order effects could have contributed to image type effects for these two variables, so we exclude them from further consideration in this paper and focus instead on the other three variables.

Image Type: For the MANOVA, the main effect of image type was significant for the SDT measures, $F(10, 320) = 20.44$, $p < .01$. Post-hoc paired-sample t -tests on d' , A' , $hr\text{-}far$ were also significant, t 's(340) > 2.14, p 's < .05². The values of the three SDT measures are largest for the original image-type, and tended to decrease for grayscale images, and were lowest for reflectance images (see Fig. 4). This indicates that for our participants both sensitivity and accuracy were highest on original color images, moderate for grayscale images, and lowest on reflectance images. This means that both color (removed in grayscale images) and shading (removed in reflectance images) are important for discriminating CG images and photos, but shading is more important than color for visual realism evaluation.

Experts and Laypersons: Classification accuracy, calculated on each image, is the percent across participants the image is correctly classified (see Fig. 5). An ANOVA on classification accuracy showed an interaction between image category (CG or photo) and expertise-level, $F(1, 58) = 23.52$, $p < .001$. Across image-types, there was better expert relative to layperson impact on classification accuracy for photos, t 's(9) > 3.9, p 's < .01, but not for CG images, $|t$'s(9) < 1.07, p 's > .311. Perhaps experiences in experts' daily work, such as exposure to retouched photos, helps them detect photo images, and causes their tendency to mistake some CG images for retouched photos (note that expert participants from animation corporations were excluded). Prior knowledge and experience thus affects accuracy of visual realism.

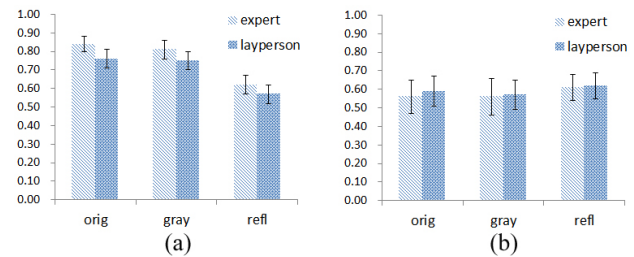


Figure 5: Average classification accuracy on photo (a) and CG (b) of three image types.

A MANOVA for expertise-level found a marginal effect on the SDT variables, $F(5, 325) = 2.04$, $p = .073$. The follow-up tests indicated better performance by experts than laypersons on d' and $hr\text{-}far$ on grayscale images only, $|t$'s(339) > 2.22, p 's < .05 (see Fig. 4). This suggests that under the loss of color the performance of laypersons drops more than that of experts. Therefore color seems to be a more important factor for visual realism for laypersons than experts. For gender effect, MANOVA showed no main or interaction effects are significant, $F(5, 313) = 1.23$, $p = .295$.

Participant Self-report of Cues Used: When indicating which cue was the most important for judgments of whether an image was CG or a photo, the largest percent (36%) of both experts and laypersons selected eyes, followed by skin (22%), illumination (17%), expression (11%), and then color (2%) (see Fig. 6). Future research will be required to validate these findings given that psychologists have found that people often fail to understand what influences their

value, $p = p$ value". If a p value is smaller than the conventional significance level threshold of .05, we reject the null hypothesis of no difference between the means.

²We report the results of t -tests as, " $t(df) = t$ value, $p = p$ value".

¹We report the results of ANOVAs as, " $F(df_{condition}, df_{error}) = F$

judgments and perceptions [Nisbett and Wilson 1977]. Nonetheless, these findings may provide insights into what attributes are important to include in a computational model of visual realism.

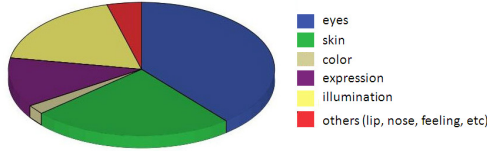


Figure 6: Main clues for judgment claimed by participants.

3 Discussion

The goal of this work was to explore factors of images and characteristics of human judges that affect the perception of visual realism for face images. To achieve this goal we combined the novel approach of intrinsic image decomposition with experimental methods and signal detection theory. One key finding was that participants discriminated between photos and CG images best for original images, less well for grayscale images, and worst for reflectance images. This suggests that shading may be more important than color for judgments about visual realism. Another key finding was that experts outperformed laypersons in discriminating tasks (see Fig. 4), but surprisingly only for grayscale images. This finding suggests that experts better utilize shading information than do laypersons, but not color information. This suggests a tip for CG artists: if your audience consists of experts be especially careful with your image’s shading, but if your audience is laypersons then pay more attention to the image’s color. For any audience, however, shading is important. Our findings suggest some additional research questions. For example, is using shading information more amenable to expertise-related training than using color information? How much shading realism is required for a CG image to fool human judges? Will our results for faces generalize to image content such as whole bodies or inanimate objects? Finally, since lighting and other artifacts are sensitive to image size [Larkin and Sullivan 2011], will our conclusions generalize to images of other sizes?

Besides the work described in the paper, we also repeated the experiment in [Farid and Bravo 2012]’s dataset on original components (see Fig. 2(b)), trying to evaluate the impact of cognitive factors on visual realism. We compare the performance of our Chinese participants with that in [Farid and Bravo 2012] (whose participants are presumed to be predominately westerners). We got two preliminary observations: ethnicity is a strong factor for perceived realism—people have higher sensitivity on faces of their own races; details on skin texture (wrinkles, freckles, etc) may be more effective in inducing realism perception among audience of laypersons, while experts focus more on skin glossiness. We will explore these observations in our future research.

4 Conclusions

Our paper makes four main contributions. *First* is a *CG-or-Photo* face dataset that provides a common platform to analyze human perception of visual realism for face images. *Second*, it reports a set of human describable attributes of image realism that offers novel insights into mid-level feature design for realism evaluation algorithms. *Third*, it proposes a new method of image realism evaluation using intrinsic image decomposition. *Finally*, it demonstrates that cognitive factors influence human perception of image realism, which is potentially important for the animation industry, news media, and forensics investigation.

We hope that our work inspires the computer graphics community and psychologists to conduct additional research aimed at understanding the factors that affect the visual realism of face images, thus enabling improvement in CG image evaluation. Such research may also help computer graphics designers to produce more realistic depictions of faces by directly targeting the cues that their audience uses to detect that an image is CG. We also hope our findings are helpful for researchers in the field of image forensics as they seek more intelligent algorithms to distinguish CG images from photos.

Acknowledgements

We thank David Pautler and other psychologists from IHPC, A*STAR, and Prof. Fu of Zhejiang Gongshang University for helpful discussions on the paper. We thank Hui-Cheng Tan and Annie Koh from National University of Singapore for their help on the psychophysical experiment arrangement. We also want to thank Hany Farid for sharing his face dataset, and Roger Grosse for sharing the code of intrinsic image decomposition. Finally, we thank photographers from photoSIG, Jean Pierre, Oscar. E.V. Tod, Yuri, and other anonymous photographers, for providing great portrait images for our dataset. This work is supported by Research Project of Department of Education in Zhejiang Province, China (No. Y201016486).

References

- BAILEY, R. A. 2008. *Design of comparative experiments*. Cambridge University Press.
- BRUCE, V., AND YOUNG, A. 2012. *Face Perception*, 1 ed. Psychology Press.
- FARID, H., AND BRAVO, M. J. 2012. Perceptual discrimination of computer generated and photographic faces. *Digital Investigation* 4, 226–235.
- GROSSE, R., JOHNSON, M. K., ADELSON, E. H., AND FREEMAN, W. T. 2009. Ground truth dataset and baseline evaluations for intrinsic image algorithms. In *IEEE International Conference on Computer Vision*.
- MCMANARA, A. 2005. Exploring perceptual equivalence between real and simulated imagery. In *Proceedings of the 2nd symposium on APGV*, vol. 38.
- LARKIN, M., AND SULLIVAN, C. O. 2011. Perception of simplification artifacts for animated characters. In *Proceedings of ACM SIGGRAPH symposium on APGV*, 93–100.
- MEYER, G. W., RUSHMERE, H. E., COHEN, M. F., GREENBERG, D. P., AND TORRANCE, K. E. 1986. An experimental evaluation of computer graphics imagery. *ACM Transactions on Graphics* 5, 1, 30–50.
- MCDONNELL, R., BREIDT, B., BULTHOFF, H. H. 2012. Render me real? Investigating the effect of render style on the perception of animated virtual humans. *ACM Transactions on Graphics*.
- NG, T. T., AND CHANG, S. F. 2012. Discrimination of computer synthesized or recaptured images from real images. In *Digital Image Forensics*, Springer, 275–309.
- NISBETT, R. E., AND WILSON, T. D. 1977. Telling more than we can know. *Psychological Review* 84, 231–259.
- WICKENS, D. T. 2002. *Elementary signal detection theory*. Oxford University Press.